

Where should Brazil open its next neonatal intensive care units?

Access of very-low-birth-weight births, 2006–2025, and a maximal covering location model

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Abstract

A newborn of 500 to 1,499 g depends on neonatal intensive care from the first hours of life. This note measures, from Brazilian open data, how access to the neonatal intensive care units (NICUs) of the public health system (SUS) changed over the 20 years from 2006 to 2025 for the 693,038 live births in this weight range, whether the units opened over the period went where births were out of reach, and where new units would bring the most births within reach. Travel time is measured from the municipality of residence on the IBGE survey of bus and boat lines. The share of these births born in a facility with a SUS NICU rose from 47.7 % to 67.3 %, and the share living more than two hours from one fell from 14.0 % to 9.3 %, most of it by 2010. The North did not follow, and 29.7 % of its births at risk still live more than two hours away. Of the 112 municipalities that opened their first SUS NICU, 34 were more than two hours from any unit. A maximal covering location model, solved exactly on the births of 2021–2025, finds that ten new units would bring 20 % of the uncovered births within two hours, and that no choice among the 217 candidate municipalities could reach more than 64 %. The sites chosen in at least four of the five years and under straight-line distances are Arcoverde (PE) and Jacobina (BA).

Resumo

Um recém-nascido de 500 a 1.499 g depende de terapia intensiva neonatal desde as primeiras horas de vida. Esta nota mede, com dados abertos brasileiros, como evoluiu de 2006 a 2025 o acesso dos 693.038 nascidos vivos nessa faixa de peso às UTIs neonatais do SUS, se as unidades abertas no período foram para onde os nascimentos estavam fora de alcance e onde novas unidades trariam mais nascimentos para perto de uma UTI. A parcela nascida em estabelecimento com UTI neonatal SUS passou de 47,7 % para 67,3 %, e a parcela que mora a mais de duas horas de uma UTI caiu de 14,0 % para 9,3 %, sobretudo até 2010. A região Norte não acompanhou, com 29,7 % dos nascimentos de risco ainda a mais de duas horas. Um modelo de cobertura máxima, resolvido de forma exata, indica que dez novas unidades cobririam 20 % dos nascimentos descobertos em 2021–2025 e que nenhuma escolha entre os municípios candidatos passaria de 64 %.

1 Context

A newborn under 1,500 g needs intensive care from the first hours of life. Being born in a hospital that has a neonatal intensive care unit, rather than transferred after birth, is the condition on which the rest of the care builds. Brazil added NICU beds over the last twenty years. Where these beds went matters as much as how many there are, because a unit serves the births within a few hours of it and no others.

Describing the unequal distribution of units is a first step. This note asks the question a planner faces next. Access is measured for every year from 2006 to 2025, the units opened over the period are set against the births that were out of reach, and a location model indicates where the next units would bring the most births within reach, with a check of how stable that answer is from one year to the next. The note measures and ranks options under explicit assumptions. It does not assess any public policy.

2 Data

Three open sources are used. Birth records are de-identified microdata, and only counts are published.

The live birth information system (SINASC, DATASUS) records every live birth with its weight, gestational age, municipality of residence, municipality of birth and the facility where it took place. The series runs from 2006 to 2025, final up to 2024 and preliminary for 2025, with 56,162,606 births in all.

The national register of health facilities (CNES, DATASUS) gives each month the beds of every facility by type, with the number available to SUS patients. Beds are read once a year, in December, and in November for 2007 (Section 3.2).

IBGE provides the survey of regular intercity bus and boat lines of 2016 (*Ligações rodoviárias e hidroviárias*), with the minimum travel time between linked municipalities, the population arrangements, groups of municipalities that form one urban area, one point per municipality, its seat, from the 2022 locality file, and the state boundaries drawn on the map.

Raw files come from the PySUS mirror when it holds them, otherwise from the DATASUS FTP server. Where a file was read from both, the two copies were identical. Each of the 1,084 raw files is recorded with its source, row count and content hash. Birth weight is missing for 10,394 births in 2006 and 204 in 2025. These births are counted and left out, never imputed.

3 Method

3.1 Births at risk

A birth at risk is a live birth of 500 to 1,499 g. Weight is recorded the same way over the whole period, which gestational age is not. The birth form changed in 2011, and the way gestational age is recorded with it, which breaks the series of preterm births that year. Births under 500 g are counted apart. They went from 2,625 in 2006 to 3,942 in 2025 while births fell by a sixth, a rise that follows how births at the limit of viability are registered rather than a rise in need. Counted with the others they would read as growing demand. Births of 500 to 1,499 g, 693,038 over the period, follow the number of births. As a check, births under 32 weeks of 1,500 g or more are followed apart from 2011.

3.2 Neonatal intensive care beds

NICU beds are the CNES codes for neonatal intensive care, a single code until 2008 and three codes by type of unit afterwards. When a facility reports both in the same month, the larger count is kept. The main measure counts the beds available to SUS, on which most of the population depends, and a facility offers SUS neonatal intensive care when it has at least one such bed. In December 2007 SUS beds leave the old code before reappearing under the new ones, so 2007 is read in November. Every bed, SUS or not, is used as a check from 2008, the first year in which the register counts type I units as neonatal intensive care. SUS NICU beds went from 2,874 in 2006 to 5,186 in 2025, and all NICU beds from 7,182 in 2008 to 10,229.

3.3 Travel time

Distance is measured as the travel time of a family without a car. The IBGE survey gives, for every pair of municipalities linked by a regular bus or boat line, the minimum travel time, and the shortest path over this network gives a time between any two municipalities. The survey leaves

out lines inside urban areas, so every pair of municipalities of the same population arrangement is linked at 60 minutes, with 30 and 90 minutes tested. Municipalities missing from the survey, most of them without public transport, are attached to their three nearest surveyed neighbours by straight-line distance, converted into time at the median pace of the surveyed lines, 1.39 minutes per kilometre or about 43 km/h. The completed network holds the 5,571 municipalities in one piece, with 68,332 links, 378 of them estimated. A municipality is reduced to its seat and a birth to its municipality of residence. Straight-line distance at the same pace is run as a check for the whole study.

3.4 Indicators

For each year and bed scope, three indicators are computed for the country, its regions and its states. The first is the share of births at risk born in a facility with a NICU. The second is the share living in a municipality more than 90, 120 or 180 minutes from the nearest municipality with a NICU, two hours being the main threshold. The third is the share born outside the municipality of residence.

3.5 Decomposition of the change

Let $s(b, m)$ be the share of the births at risk of year b living more than two hours from the units of year m , and f and l the first and last years. The change $\Delta = s(l, l) - s(f, f)$ splits exactly into a beds effect B and a births effect N ,

$$B = \frac{1}{2}[s(f, l) - s(f, f) + s(l, l) - s(l, f)], \quad N = \frac{1}{2}[s(l, f) - s(f, f) + s(l, l) - s(f, l)],$$

so that $\Delta = B + N$. The beds effect moves the map of units with the births held still, the births effect moves the births with the map held still, each averaged over the two possible orders.

3.6 Location model

The maximal covering location problem chooses p sites among candidates so as to bring the largest number of uncovered births within a travel time T ,

$$\max \sum_i w_i y_i \quad \text{subject to} \quad y_i \leq \sum_{j \in N_i} x_j, \quad \sum_j x_j = p, \quad x_j \in \{0, 1\}, \quad 0 \leq y_i \leq 1,$$

where w_i is the number of births at risk living in municipality i more than T from any SUS NICU, N_i the candidates within T of i , x_j the choice of candidate j and y_i the coverage of i . It is solved exactly as an integer program with the HiGHS solver through SciPy, for $p = 5, 10$ and 20 and $T = 90, 120$ and 180 minutes. A candidate is a municipality without a SUS NICU where at least 1,000 births a year already take place, between 197 and 254 municipalities in each year of 2021–2025 and 217 on the five years together. A unit needs a maternity of some size, and the model should not propose a site without obstetric activity.

One year holds too few births per municipality for a stable answer. The model is therefore solved on the births at risk of 2021–2025 summed together, against the map of units of 2025. Each year is then solved alone, and for every site of the answer the note reports in how many years it is also chosen, whether it is still chosen at 90 and at 180 minutes, and whether it is still chosen under straight-line distances.

Table 1: Births of 500 to 1,499 g by region, share born in a facility with a SUS NICU and share living more than two hours from one, 2006 and 2025 (2025 preliminary).

Region	Births 500–1,499 g		Born in a NICU facility		Living over 2 hours away	
	2006	2025	2006	2025	2006	2025
Norte	2,733	3,128	46.4 %	62.1 %	37.3 %	29.7 %
Nordeste	9,289	8,158	44.3 %	68.0 %	23.7 %	13.9 %
Sudeste	15,310	12,775	55.3 %	65.0 %	4.7 %	2.2 %
Sul	4,521	4,397	31.5 %	78.6 %	5.8 %	2.7 %
Centro-Oeste	2,318	2,693	44.0 %	63.1 %	25.3 %	15.9 %
<i>Brazil</i>	34,171	31,151	47.7 %	67.3 %	14.0 %	9.3 %

4 Results

4.1 Access over twenty years

Births of 500 to 1,499 g went from 34,171 in 2006 to 31,151 in 2025. The share born in a facility with a SUS NICU rose from 47.7 % to 59.0 % in 2010 and 67.3 % in 2025 (figure 1). The share living more than two hours from a SUS NICU fell from 14.0 % (4,779 births) to 9.9 % in 2010, then barely moved, to 9.3 % (2,893 births) in 2025. At 90 minutes the share is 15.5 % in 2025, at three hours 3.8 %. The peak of 2007 coincides with the change of bed codes in the register and is read with caution. Families travel more to give birth, and 35 % of these births took place outside the municipality of residence in 2006 against 45 % in 2025.

Regions differ more than they converge (table 1). The North remains the region farthest from a unit, with 37.3 % of its births at risk more than two hours away in 2006, 28.3 % in 2010 and 29.7 % in 2025. It holds 10 % of the births at risk of 2025 and 32 % of those more than two hours away. The Northeast, with 1,137 births beyond two hours, holds the largest number.

Counting every NICU bed, SUS or not, the share born in a facility with a NICU goes from 71.0 % in 2008 to 85.9 % in 2025, and the share beyond two hours from 10.1 % to 6.1 %. Private units close part of the gap for the families who can use them.

The time given to a trip inside an urban arrangement hardly matters. Linking its municipalities at 30 or 90 minutes instead of 60 moves the share beyond two hours in 2025 by at most 0.2 percentage points. The definition of a birth at risk matters more. Births under 32 weeks of 1,500 g or more, 14,192 in 2025, are farther from units than the main group, 16.8 % of them more than two hours away against 9.3 %, so the main measure is a floor.

4.2 Where the beds went

SUS NICU beds grew by 2,312 between 2006 and 2025. Of these, 1,166 were added in municipalities that already had a unit and 1,146 in the 112 municipalities that opened their first one, while 25 municipalities lost all their SUS NICU beds (110 beds). Municipalities with a SUS NICU went from 192 to 279.

Of the 112 municipalities that opened their first unit, 34 were more than two hours from any unit in 2006, with 341 beds in 2025. The others opened within reach of an existing unit. Held at their 2006 distribution, the 4,779 births at risk that lived more than two hours away in 2006 would number 2,663 with the units of 2025 (7.8 %). The new units brought 2,423 of them within reach (51 %), and 307 births of covered municipalities fell out of reach where units closed.

The share beyond two hours changed by -4.7 percentage points between 2006 and 2025.

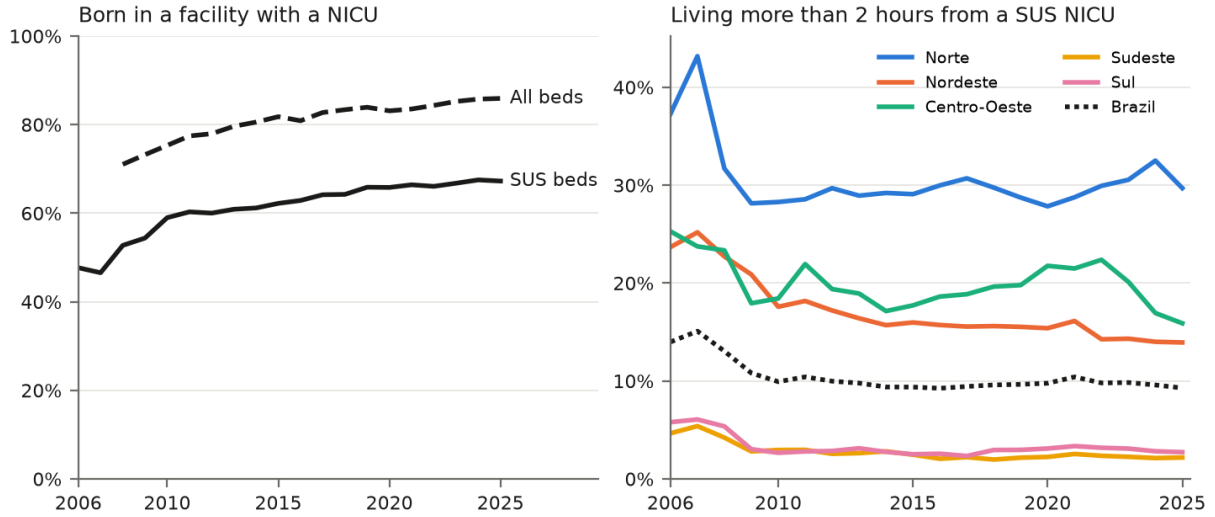


Figure 1: Births of 500 to 1,499 g, 2006–2025. Left, share born in a facility with a NICU, counting SUS beds and, from 2008, every bed. Right, share living more than two hours from a SUS NICU by region. 2007 is read in November, 2025 is preliminary.

The new map of units accounts for -7.1 points and the change in where births take place for $+2.4$ points. Births at risk moved towards regions farther from units, the North holding 8.0 % of them in 2006 and 10.0 % in 2025, so part of what the new units gained was taken back by the geography of births. Counting every bed from 2008, the change is -4.0 points, -5.3 from beds and $+1.3$ from births.

4.3 Where to open the next units

Over 2021–2025, 15,398 births at risk lived more than two hours from a SUS NICU, about 3,080 a year. Five new units would bring 1,819 of them within two hours (12 %), ten 3,010 (20 %) and twenty 4,790 (31 %). A unit in every one of the 217 candidates would cover 9,907 (64 %). The other 5,491 births live too far from any municipality large enough to hold a unit, and new units alone will not reach them. Ten units cover 16 % of the uncovered births at 90 minutes, 32 % at three hours and 20 % under straight-line distances.

Table 2 lists the ten sites and figure 2 places them. Arcoverde (PE) comes first, with 106 births at risk a year that live beyond two hours today and would be within reach. The choices differ in how solid they are. Arcoverde (PE) and Jacobina (BA) are chosen in at least four of the five years taken alone and under straight-line distances. Rolim de Moura (RO) and Crateús (CE) are chosen in at most one year alone, and the pooled answer rests on a few births that come and go. Only two of the ten sites are in the North, which holds 31 % of the uncovered births of 2021–2025. Its municipalities are far apart and each candidate reaches few births, so a model that maximises the number of births covered favours the denser Northeast. The model counts births and does not weigh equity between regions, and the gap of the North is not one that fixed units within two hours can close.

5 Limits

The transport network dates from 2016 and is applied to the whole period. Travel times are those of scheduled public transport, slower than a car or an ambulance and closer to what most families face. A municipality is reduced to its seat, so distances inside a municipality count as zero, which flatters the large municipalities of the North. SINASC records where a child was

Table 2: The ten sites of the location model, births of 2021–2025 against the SUS units of 2025. Births reached are births at risk a year, living more than two hours from any SUS NICU, that the site would bring within two hours. Years chosen counts the years, out of five, in which the model solved on that year alone also chooses the site.

Municipality	State	Births reached a year	Years chosen	Also chosen at		Straight line
				90 min	180 min	
Arcoverde	PE	106	4 of 5			•
Jacobina	BA	74	5 of 5	•		•
Rolim de Moura	RO	66	1 of 5	•		
Antas	BA	61	3 of 5	•		
Santo Antônio de Jesus	BA	58	4 of 5	•		
Juruti	PA	54	3 of 5			
Eunápolis	BA	51	3 of 5	•		•
Rio das Ostras	RJ	45	2 of 5	•		•
Barra do Corda	MA	45	3 of 5		•	
Crateús	CE	43	1 of 5			

born, not where it was treated, and transfers after birth would need hospital admission records. Beds are those declared in the register, once a year, without their occupancy, staffing or level of care. Weight leaves out preterm births of normal weight that also need intensive care, so demand is a floor. Data for 2025 are preliminary. The location model assumes that a new unit serves every birth within the travel time, whatever its size, and says nothing of cost. It maximises the number of births covered and does not weigh equity between regions.

6 Reproducibility

The pipeline is a Python command line, `nicu`. `nicu ingest` fetches the raw files and records each one in a manifest, `nicu build` reduces them with DuckDB to counts by year, municipality and facility, `nicu analyze` answers the three questions and `nicu report` writes every number, table and figure of this note. Some sentences state a fact about the data rather than a number, such as the North remaining the region farthest from a unit, and `nicu report` checks each of them and stops if the data no longer supports it. The counts and results are versioned, so the analysis and this note can be rebuilt from a clone without downloading the raw files. Continuous integration runs the tests, rebuilds the numbers of the note from the versioned results, checks that they match the committed ones and compiles the note. Design choices are recorded as decision records in the repository, github.com/paulianne-fontoura/nicu-access-brazil, and the code is under the MIT licence.

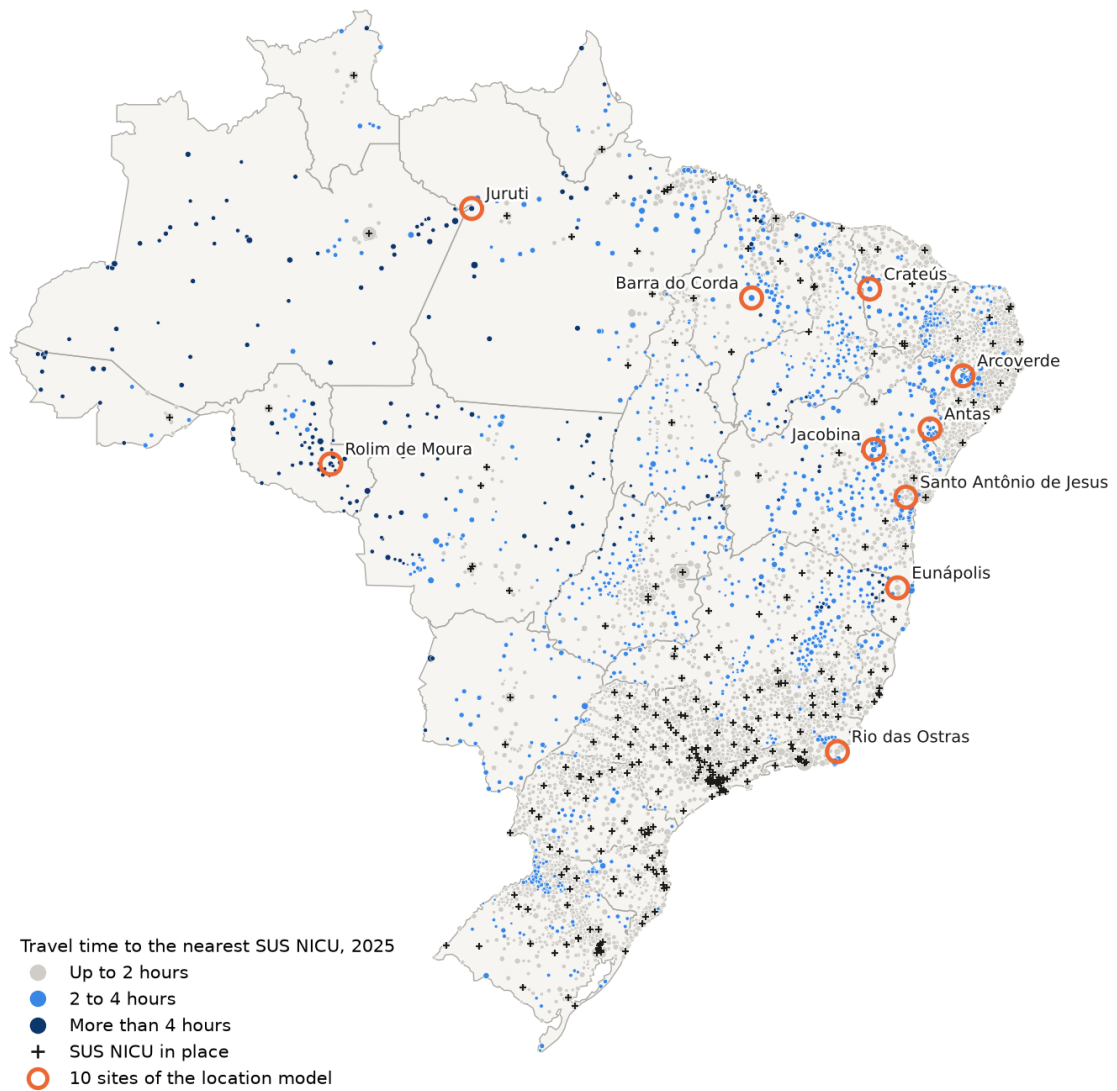


Figure 2: Travel time from each municipality to the nearest SUS NICU in 2025 on the bus and boat network. Dot size follows the births of 500 to 1,499 g of 2021–2025 by municipality of residence. Crosses mark the municipalities with a SUS NICU, circles the ten sites of the location model at two hours.